

UNIT & MEASUREMENTS

1. Measurement of large distance is done by :
 - (1) Metre scale method
 - (2) Parallax method
 - (3) Distance method
 - (4) Position method
2. Suppose true value of certain length is near 3.578 cm student took three reading with different scale.
 - (a) 3.5 cm using 0.1 cm resolution scale
 - (b) 3.38 cm using 0.01 cm resolution scale
 - (c) 3.372 cm using 0.001 cm resolution scale then
 - (1) First is more accurate
 - (2) 2nd is more accurate
 - (3) 3rd is more precise
 - (4) Both (1) and (3) are correct
3. Which error is associated with the resolution of the instrument
 - (1) Systematic error (2) Random error
 - (3) Least count error (4) All of these
4. Correct increasing order of time intervals.
 - (a) Rotation period of Earth
 - (b) Age of Egyptian pyramid
 - (c) Average human Life-span
 - (d) Travel time for light from Sun to the earth.
 - (1) $a > b > c > d$ (2) $b > a > c > d$
 - (3) $c < a < d < b$ (4) $d < a < c < b$
5. An optical microscope cannot resolve particles with size smaller than
 - (1) 10000 Å (2) 8000 Å
 - (3) 6000 Å (4) 3000 Å
6. Select the incorrect are :
 - (1) 0.06900 has 4 number of significant figures.
 - (2) 0.6900 has 4 number of significant figures.
 - (3) 6.900 has 2 number of significant figures.
 - (4) 6.900 kg has 4 number of significant figures.
7. Select the number of correct statement
 - (a) One astronomical unit (1AU) is the maximum distance of Sun from the earth.
 - (b) One parsec is the distance at which average radius of the Earth's orbit subtends an angle of 1 arc second.
 - (c) 1 a.m.u. is $\frac{1}{12}$ th of mass of an atom of carbon-12 isotope.
 - (d) Spherometer can be used to measure angle of spherical bodies.
 - (1) One (2) Two
 - (3) Three (4) Four
8. Which of the following measurement is most precise ?
 - (1) 6.00 mm (2) 6.00 cm
 - (3) 6.00 m (4) 6.00 km
9. Device used for measuring the mass of atoms and molecules.
 - (1) Chronometer (2) Spectrometer
 - (3) Spectroscope (4) Anemometer
10. Which of the following time measuring devices is most precise ?
 - (1) A wall clock (2) a stop watch
 - (3) A digital watch (4) an atomic clock
11. Match the column

Column-I

 - (p) Quantity has unit but no dimensions
 - (q) Quantity has neither unit nor dimensions
 - (r) A constant which has a unit.

Column-II

 - (a) Relative density
 - (b) Solid angle
 - (c) Reynolds number
 - (d) permittivity of free space
 - (1) p-a; q-b; r-c
 - (2) p-c; q-d; r-b
 - (3) p-d; q-c; r-a
 - (4) p-b; q-a, c; r-d

12. We have different unit for same physical quantity because same physical quantity
- (1) is measured by different instrument
 - (2) is measured by different method
 - (3) is measured with the different significant number
 - (4) is measured with the different orders of the quantity
13. The previous definition of the SI unit of one kilogram is equal to the mass of the international prototype of the kilogram i.e.,
- (1) a platinum - iridium alloy sphere
 - (2) a platinum - cylinder
 - (3) a iridium sphere
 - (4) a platinum - iridium alloy cylinder
14. Which type of error influence the accuracy of a large number of measurement result
- (1) Random error
 - (2) Systematic error
 - (3) Both
 - (4) None
15. Dimensionally which of the following equation is not correct :-
(Here a = amplitude, t = time, T = time period, ω = angular velocity, x = displacement)
- (1) $x = a \sin \left(\frac{2\pi t}{T} \right)$
 - (2) $x = a \cos (\omega t)$
 - (3) $x = \frac{a}{T} \sin \left(\frac{t}{a} \right)$
 - (4) $x = a\sqrt{2} \left(\sin \frac{2\pi t}{T} - \cos \frac{2\pi t}{T} \right)$
16. Light year is
- (1) Light emitted by the Sun in one year
 - (2) The time taken by light to travel from Sun to earth
 - (3) The distance travelled by light in free space in one year
 - (4) The time taken by earth to go once around the Sun
17. Which of the following statements is incorrect
- (1) Direct and indirect methods are used for the measurement of physical quantities.
 - (2) Scientific notation and the prefixes are used to simplify numerical computation
 - (3) A dimensionally correct equation need not to be a correct equation
 - (4) The SI units is based on six base units
18. If power (P), Surface tension (S) and Planck's constant (h) are arranged so that the dimensions of time in their dimensional formulae are in ascending order then which of the following is correct ?
- (1) P, S, h
 - (2) P, h, S
 - (3) S, P, h
 - (4) S, h, P
19. Checking the correctness of equation using the method of dimensions is based on
- (1) The type of system
 - (2) Equality of inertial frames of references
 - (3) Principle of homogeneity of dimensions
 - (4) None of these
20. The best method of reducing random error is
- (1) to change the instrument used for measurement
 - (2) to take help of experienced observer
 - (3) to repeat the experiment many times and to take the average results
 - (4) None of the above
21. Which of the following is the most precise instrument for measuring length
- (1) Metre rod of least count 0.1 cm
 - (2) Vernier callipers of least count 0.01 cm
 - (3) Screw gauge of least count 0.001 cm
 - (4) Data is not sufficient to decide.

22. From the equation $\tan \theta = \frac{rg}{v^2}$, one can obtain the angle of banking θ (from vertical) for a cyclist taking a curve (the symbols have their usual meanings). The equation is

- (1) both dimensionally and numerically correct
- (2) neither numerically nor dimensionally correct
- (3) dimensionally correct only
- (4) numerically correct only

23. Check the correctness of the relation

$S = ut + \frac{1}{2}at^2$, where u is initial velocity, a is acceleration, t is time and S is displacement

- (1) Dimensionally correct
- (2) Dimensionally incorrect
- (3) Can't explain
- (4) Data insufficient

24. Write the dimensions of a in the relation,

$$P = \frac{b - x^2}{at}$$

where P is power, x the distance and t the time.

- (1) $[M^0L^0T^0]$
- (2) $[M^1L^0T^2]$
- (3) $[M^{-1}L^0T^2]$
- (4) $[M^0LT^{-1}]$

25. The centripetal force F acting on a particle moving uniformly in a circle may depend upon mass (m), velocity (v) and radius (r) of the circle. Derive the formula for F using the method of dimensions

$$(1) F = \frac{mv}{r^2} \quad (2) F = \frac{mv^2}{r}$$

$$(3) F = \frac{mr}{v} \quad (4) F = \frac{mv}{v}$$

26. Add 6.75×10^3 cm and 4.52×10^2 cm with regard to significant figures

- (1) 7.20×10^3 cm
- (2) 7.20×10^2 cm
- (3) 7.20×10^4 cm
- (4) 720 cm

27. A thin wire has a length of 21.7 cm and radius 0.46 mm. Calculate the volume of the wire to correct significant figures.

- (1) 0.14 cm^3
- (2) 0.144 cm^3
- (3) 0.014 cm^3
- (4) 0.143 cm^3

28. Arrange the following in the most correct decreasing order of preciseness for measuring length ?

- (a) A vernier calliper with 20 divisions on the sliding scale coinciding with 19 main scale divisions of main scale
- (b) A screw gauge of pitch 1 mm and 100 divisions on the circular scale
- (c) An optical instrument that can measure length to within a wavelength of light

- (1) c, a, b
- (2) c, b, a
- (3) a, b, c
- (4) b, c, a